

StringOperations.java — Code Explanation (Updated Version)

1. Program in Simple Words

This program is a **menu-driven String operations demo**. It takes three strings (`s1`, `s2`, `s3`) as input from the user, then repeatedly shows a menu of 15 common String operations (concatenation, comparison, searching, replacing, splitting, extracting a substring, formatting, and reference vs. content equality). The user picks a number, the matching operation runs on `s1`, `s2`, and/or `s3`, and the result is printed. This repeats in a loop until the user chooses option 16 to exit.

Note on changes from the earlier version:

- Option 11 now runs `s2.replace(s1, s3)` (replaces `s1` inside `s2`) instead of `s1.replace(s2, s3)`.
- Option 13 is now `substring()` — it asks the user for a start and end index and extracts that portion of `s1`, replacing the earlier `StringBuilder` demo.

2. Step-wise Execution

1. Scanner object is created to read input.
2. User enters three strings: `s1`, `s2`, `s3`.
3. `showMenu()` prints the list of 16 options (15 operations + Exit).
4. Program enters a do-while loop:
 - a. User enters a choice number.
 - b. `scanner.nextLine()` clears the leftover newline after `nextInt()`.
 - c. A switch statement checks the choice:
 - 1 -> concatenates `s1 + s2` using `'+'`
 - 2 -> concatenates `s1` and `s2` using `concat()`
 - 3 -> joins `s1`, `s2`, `s3` with `","` using `String.join()`
 - 4 -> compares `s1` and `s2` using `equals()`
 - 5 -> compares `s1` and `s2` ignoring case using `equalsIgnoreCase()`
 - 6 -> compares `s1` and `s2` lexicographically using `compareTo()`
 - 7 -> checks if `s1` contains `s2` using `contains()`
 - 8 -> checks if `s1` starts with `s2` using `startsWith()`
 - 9 -> checks if `s1` ends with `s2` using `endsWith()`
 - 10 -> finds position of `s2` inside `s1` using `indexOf()`
 - 11 -> replaces occurrences of `s1` in `s2` with `s3` using `replace()`
 - 12 -> joins `s1`, `s2`, `s3` with `","` then splits them back using `split()`
 - 13 -> reads start and end positions from user, extracts `s1.substring(start, end)` and prints it
 - 14 -> formats `s1`, `s2`, `s3` as `"s1-s2-s3"` using `String.format()`
 - 15 -> compares `s1` and `s2` using `==` (reference) and `equals()` (content)
 - 16 -> prints "Program ended." and loop condition becomes false
 - default -> prints "Invalid choice."

- d. Loop repeats until choice == 16.
5. Scanner is closed.

3. Algorithm

START

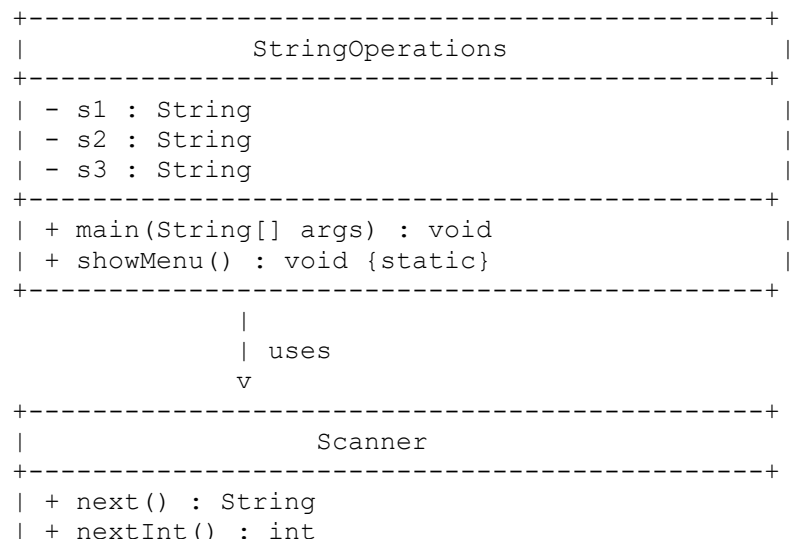
1. Create Scanner object
2. Read s1, s2, s3 from user
3. Display menu of operations
4. REPEAT:
 - a. Read user's choice
 - b. SWITCH on choice:
 - CASE 1: Print s1 + " " + s2
 - CASE 2: Print s1.concat(s2)
 - CASE 3: Print String.join(" ", s1, s2, s3)
 - CASE 4: Print s1.equals(s2)
 - CASE 5: Print s1.equalsIgnoreCase(s2)
 - CASE 6: Print s1.compareTo(s2)
 - CASE 7: Print s1.contains(s2)
 - CASE 8: Print s1.startsWith(s2)
 - CASE 9: Print s1.endsWith(s2)
 - CASE 10: Print s1.indexOf(s2)
 - CASE 11: Print s2.replace(s1, s3)
 - CASE 12: Join s1,s2,s3 with "," then split and print array
 - CASE 13: Read start, end from user
Print s1.substring(start, end)
 - CASE 14: Print String.format("%s-%s-%s", s1, s2, s3)
 - CASE 15: Print (s1 == s2) and s1.equals(s2)
 - CASE 16: Print "Program ended."
 - DEFAULT: Print "Invalid choice."

UNTIL choice == 16

5. Close Scanner

STOP

4. UML Class Diagram



```

| + nextLine() : String
| + close() : void
+-----+

+-----+
|                               |
|                               |
+-----+
| + concat(String) : String
| + equals(Object) : boolean
| + equalsIgnoreCase(String) : boolean
| + compareTo(String) : int
| + contains(CharSequence) : boolean
| + startsWith(String) : boolean
| + endsWith(String) : boolean
| + indexOf(String) : int
| + replace(CharSequence, CharSequence) : String
| + split(String) : String[]
| + substring(int, int) : String
| + join(CharSequence, CharSequence...) : String {static}
| + format(String, Object...) : String {static}
|
+-----+

```

Code:

```

package basics;

import java.util.Scanner;
import java.util.Arrays;

public class StringOperations {
    static void showMenu() {
        System.out.println("\nChoose a string operation:");
        System.out.println("1. Concatenate using +");
        System.out.println("2. Concatenate using concat()");
        System.out.println("3. Join strings with
String.join()");
        System.out.println("4. Compare using equals()");
        System.out.println("5. Compare using
equalsIgnoreCase()");
        System.out.println("6. Compare lexicographically using
compareTo()");
        System.out.println("7. Check contains()");
        System.out.println("8. Check startsWith()");
    }
}

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```

        System.out.println("9. Check endsWith()");
        System.out.println("10. Find position using
indexOf()");
        System.out.println("11. Replace using replace()");
        System.out.println("12. Join and split strings");
        System.out.println("13. Get substring");
        System.out.println("14. Format strings using
String.format()");
        System.out.println("15. Compare == with equals()");
        System.out.println("16. Exit");
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter s1: ");
        String s1 = scanner.next();
        System.out.print("Enter s2: ");
        String s2 = scanner.next();
        System.out.print("Enter s3: ");
        String s3 = scanner.next();

        int choice;
        showMenu();
        do {
            System.out.print("Enter your choice: ");
            choice = scanner.nextInt();
            scanner.nextLine();

            switch (choice) {
                case 1: {
                    System.out.println("Using += " + s1 + " " +
s2);
                    break;
                }
                case 2: {

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                System.out.println("Using concat(): " +
s1.concat(s2));
                break;
            }
            case 3: {
                System.out.println("Joined string: " +
String.join(", ", s1, s2, s3));
                break;
            }
            case 4: {
                System.out.println("equals(): " +
s1.equals(s2));
                break;
            }
            case 5: {
                System.out.println("equalsIgnoreCase(): " +
s1.equalsIgnoreCase(s2));
                break;
            }
            case 6: {
                System.out.println("compareTo(): " +
s1.compareTo(s2));
                break;
            }
            case 7: {
                System.out.println("Contains s2: " +
s1.contains(s2));
                break;
            }
            case 8: {
                System.out.println("Starts with s2: " +
s1.startsWith(s2));
                break;
            }
            case 9: {
```

```

        System.out.println("Ends with s2: " +
s1.endsWith(s2));
        break;
    }
    case 10: {
        System.out.println("Index of s2: " +
s1.indexOf(s2));
        break;
    }
    case 11: {
        System.out.println("Replaced string: " +
s2.replace(s1, s3));
        break;
    }
    case 12: {
        String joined = String.join(",", s1, s2,
s3);
        System.out.println("Split result: " +
Arrays.toString(joined.split(",")));
        break;
    }
    case 13: {
        System.out.print("Enter start position: ");
        int start = scanner.nextInt();
        System.out.print("Enter end position: ");
        int end = scanner.nextInt();
        System.out.println("Substring: " +
s1.substring(start, end));
        break;
    }
    case 14: {
        System.out.println("Formatted string: " +
String.format("%s-%s-%s", s1, s2, s3));
        break;
    }
    case 15: {

```

```
        System.out.println("Using ==: " + (s1 ==
s2));
        System.out.println("Using equals(): " +
s1.equals(s2));
        break;
    }
    case 16:
        System.out.println("Program ended.");
        break;
    default:
        System.out.println("Invalid choice.");
    }
    System.out.println();
} while (choice != 16);

scanner.close();
}
}
```

Output:

```
File Edit Selection View Go Run Terminal Help
basicjava
StringOperations

PS C:\Users\Admin\OneDrive\Documents\basicjava> & "C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-enable-preview" "-XX:+ShowCodeDetailsInExceptionMessages" "-cp" "C:\Users\Admin\OneDrive\Documents\basicjava\bin" "basics.StringOperations"
Enter s1: qwerty
Enter s2: bitcoin
Enter s3: qwerty

Choose a string operation:
1. Concatenate using +
2. Concatenate using concat()
3. Join strings with String.join()
4. Compare using equals()
5. Compare using equalsIgnoreCase()
6. Compare lexicographically using compareTo()
7. Check contains()
8. Check startsWith()
9. Check endsWith()
10. Find position using indexOf()
11. Replace using replace()
12. Join and split strings
13. Get substring
14. Format strings using String.format()
15. Compare == with equals()
16. Exit
Enter your choice: 1
Using +: qwerty bitcoin

Enter your choice: 2
Using concat(): qwertybitcoin

Enter your choice: 3
Joined string: qwerty, bitcoin, qwerty

Enter your choice: 4
equals(): false

Enter your choice: 5
equalsIgnoreCase(): false

Enter your choice: 6
```

```
File Edit Selection View Go Run Terminal Help
basicjava
StringOperations

Enter your choice: 6
compareTo(): 15

Enter your choice: 7
Contains s2: false

Enter your choice: 8
Starts with s2: false

Enter your choice: 9
Ends with s2: false

Enter your choice: 10
Index of s2: -1

Enter your choice: 11
Replaced string: bitcoin

Enter your choice: 12
Split result: [qwerty, bitcoin, qwerty]

Enter your choice: 13
Enter start position: 1
Enter end position: 5
Substring: wert

Enter your choice: 6
compareTo(): 15

Enter your choice: 14
Formatted string: qwerty bitcoin qwerty

Enter your choice: 15
Using ==: false
Using equals(): false

Enter your choice: 16
Program ended.
```